

Filesystems - FAT, NTFS, Refs, EXT3and EXT4-05

1. Introduction

A **filesystem** defines how data is stored, organized, and retrieved on a storage device. It manages file naming, access, permissions, and reliability. Different operating systems use different filesystems based on speed, compatibility, and data protection needs.

2. FAT (File Allocation Table)

- **Developed by:** Microsoft (late 1970s).
- **Types:** FAT12, FAT16, FAT32, exFAT.
- **Features:**
 - Simple and widely supported.
 - Compatible across Windows, Linux, macOS, cameras, and USB drives.
- **Limitations:**
 - File size limit: 4 GB (FAT32).
 - Volume size limit: 2 TB.
 - No security features (permissions, encryption).
- **Use Case:** USB drives, memory cards, and devices requiring cross-platform compatibility.



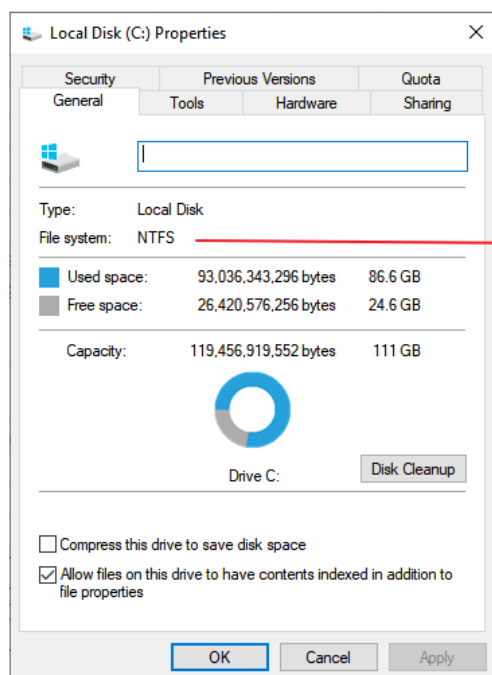
FAT File System Structure



FAT32 File System Structure

3. NTFS (New Technology File System)

- **Developed by:** Microsoft (1993, Windows NT).
- **Features:**
 - Supports large files and partitions (up to 16 TB).
 - File compression and encryption (EFS).
 - Journaling for crash recovery.
 - Access control through file permissions.
- **Limitations:**
 - Limited support in non-Windows systems (Linux/macOS need extra drivers).
- **Use Case:** Windows system drives, enterprise storage.



NTFS

New
Technology
File
System

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4. ReFS (Resilient File System)

- **Introduced by:** Microsoft in Windows Server 2012.
- **Features:**
 - Designed for data integrity and availability.
 - Auto-corrects file corruption using checksums.
 - Supports large volumes (up to 1 Yottabyte).
 - Works with Storage Spaces for redundancy.

- **Limitations:**
 - Not supported as a boot drive.
 - Some NTFS features like compression and EFS missing.
- **Use Case:** Data centers, virtualization, backup servers.

5. EXT3 (Third Extended File System)

- **Developed by:** Linux community (2001).
- **Features:**
 - First Linux filesystem with journaling.
 - Backward compatibility with EXT2.
 - Supports file size up to 2 TB.
- **Limitations:**
 - Limited scalability compared to modern filesystems.
 - Slower performance with large files.
- **Use Case:** Legacy Linux systems.

Features	Ext2	Ext3	Ext4
Individual file size	16GB-2TB	16GB-2TB	16GB-16TB
Volume file system size	4TB-32TB	4TB-32TB	4TB-1EB
Default inode size	128 bytes	128 bytes	256 bytes
Time Stamp	No support	Second	Nanosecond
Defragmentation	No	No	Yes
Directory Indexing	Disabled	Disabled	Enabled
Multiple Block Allocation	Basic	Basic	Advanced
Preallocation	No	In-core reservation	For extent file
Delayed Allocation	No	No	Yes

6. EXT4 (Fourth Extended File System)

- **Introduced:** 2008, successor to EXT3.
- **Features:**
 - Supports very large volumes (up to 1 EB).
 - Backward compatible with EXT3.

- Faster allocation using extents.
 - Improved reliability with journaling and checksums.
- **Limitations:**
 - Mostly used in Linux; limited Windows support.
- **Use Case:** Modern Linux distributions (Ubuntu, Debian, Red Hat).